

SUMMER RESEARCH INTERNSHIP PROGRAMME

2026

FINAL RESEARCH REPORT — SUBMISSION COVER PAGE

PROJECT / RESEARCH TITLE

ALAS-SCLF: Adaptive-L Asymmetric-Segmented SCLF Polar Decoders with Guard-Band Stabilization

PROGRAMME DETAILS

Domain / Track:	Communication Engineering & Wireless Channel Coding	Duration:	June 2026 – August 2026
Department:	Computer Science and Engineering	Batch / Year:	4th Year / 2026

SUBMITTED BY

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UNDER THE MENTORSHIP OF

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DECLARATION

We hereby declare that this report is a record of genuine work carried out by us under the guidance of the above-mentioned mentor, as part of the IEEE SMC Student Branch Chapter, KGEC — Research Internship Programme 2026, and has not been submitted elsewhere for any other purpose.

Mentor's Signature

Programme Coordinator

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6. Use active voice wherever possible ("We implemented the model" rather than "The model was implemented by us").
7. Define all abbreviations and technical terms at first use before using the short form.
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12. Keep sentences concise; break long, complex sentences into shorter, readable ones.
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14. Have the mentor review at least one full draft before final submission, and incorporate feedback carefully.
15. Do a final read-through as a team to check flow, coherence, and alignment between all sections before converting

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AI Usage Declaration

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5) Introduction

In modern wireless communication standards, such as 5G New Radio (NR), the requirements for Ultra-Reliable Low-Latency Communications (URLLC) mandate highly efficient channel coding schemes. Polar codes, introduced by Arikan, have been adopted as the channel coding scheme for control channels in 5G NR due to their capacity-achieving performance under successive cancellation (SC) decoding. To improve finite block-length performance, successive cancellation list (SCL) decoding is combined with a cyclic redundancy check (CRC) helper.

To bridge the performance gap between low-complexity SC and high-performance SCL with large list sizes, Successive Cancellation List Flip (SCLF) decoders restart SCL decoding by flipping error-prone bits when the initial SCL run fails the CRC. Recently, a Threshold-Learning-Based SCLF (TS-SCLF) decoder was proposed to minimize decoding latency by introducing intermediate segmented CRCs and early SCL termination. However, TS-SCLF suffers from three main limitations:

1. Post-Flip Metric Instability: Right after a bit is forcefully flipped at layer l_{flip} , the path metrics (PMs) of candidate paths undergo a severe transient period. Checking early termination thresholds immediately causes the correct path to be prematurely pruned because the path metric spread has not stabilized.
2. False-Pass Search Space Lockout: TS-SCLF uses a symmetric 50/50 partition of the information bits. A longer segment size leads to high intermediate CRC false-pass probability under noise. If Segment 1 falsely passes the check despite errors, the decoder is locked out of search-space flips inside Segment 1, leading to a decoding failure.
3. Rigid Search List Constraints: Under severe channel noise, keeping the list size restricted to $L = 4$ restricts search capacity, causing the decoder to fail to identify the correct path even with multiple flips.

6) Theoretical Background

This section establishes the mathematical foundations of polar coding, successive cancellation decoding, SCL list extension, and the Deep Q-Network (DQN) optimization framework. All formulations are generic and applicable to any codeword length $N = 2^n$ and information bits K .

6.1 Polar Codes

Polar codes polarize N independent subchannels into capacity-achieving subchannels. The polar generator matrix G_N is constructed via the Kronecker power of the kernel matrix F :

$$G_N = B_N \cdot F^{\otimes n}$$

where B_N is the $N \times N$ bit-reversal permutation matrix. The K information payload bits are mapped to the high-reliability subchannels index set I . The remaining $N - K$ subchannels represent the frozen set I^c , fixed to zero. The transmitted codeword vector is computed as $x = u \cdot G_N$.

6.2 SC Based decoding

In successive cancellation, decoding is performed recursively over the binary decoding tree. Let $P = (P_1, P_2, \dots, P_M)$ represent the LLR vector received from a parent node of length M . The left child node LLRs (P_{left}, i) and right child node LLRs (P_{right}, i) are recursively updated for i in $\{1, 2, \dots, M/2\}$ as:

$$\begin{aligned} P_{\text{left}}, i &= 2 \cdot \arctanh(\tanh(P_{2i-1}/2) \cdot \tanh(P_{2i}/2)) \\ P_{\text{right}}, i &= P_{2i} + (1 - 2u_{\text{hat_left}}, i) \cdot P_{2i-1} \end{aligned}$$

where $u_{\text{hat_left}}, i$ represents the hard-decision feedback bit computed from the left child node. At each leaf node l corresponding to an information bit ($l \in I$), the active path splits into two candidate paths representing $u_l = 0$ and $u_l = 1$. The path metric PM for path m is updated recursively according to:

$$PM_{l^{\wedge}(m)} = PM_{l-1^{\wedge}(m)} + \{ 0, \text{ if } u_{l^{\wedge}(m)} = \text{Hard}(P_{l^{\wedge}(m)}); |P_{l^{\wedge}(m)}|, \text{ otherwise } \}$$

6.3 DeepQ-Network (DQN)

To automatically discover optimal parameters, we formulate a Markov Decision Process (MDP) defined by the tuple $(S, A, P, R, \gamma_{\text{RL}})$:

1. State Space (S): The state vector represents the parameter configuration $s_t = [D_1, t, D_2, t, \alpha_t]$.
2. Action Space (A): The action space consists of discrete adjustments a_t in $\{ +\delta_{D_1}, +\delta_{D_2}, +\delta_{\alpha} \}$.
3. Reward Function (R): Evaluates error-correction performance and attempts compared to reference values:

$$R(s) = \{ T_{\text{prev}} - T_{\text{curr}}(s), \text{ if } FER(s) \leq (1 + \delta) \cdot FER_{\text{ref}}; -1000, \text{ otherwise } \}$$

7) Proposed ALAS-SCLF

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This section describes the generic, value-independent concepts of the proposed ALAS-SCLF decoder. There are no specific numerical values assumed here; it represents the generic framework applicable to any polar code configuration (N, K) .

7.1 Understanding of Work

1. Post-Flip Guard-Band (G): Flipped bits cause a transient non-stationary shift in LLR and path metrics. Checking thresholds immediately at layer $l_{\text{flip}} + 1$ prunes correct paths. We disable early termination threshold checks for a Guard-Band window G defined as:

$$\text{Early Termination Active iff: } l > l_{\text{flip}} + G, \quad G = \text{ceil}(\beta \cdot N), \quad \beta \approx 0.06$$

2. Asymmetric Partitioning & Segment-Constrained Flipping (SCF): The payload K is partitioned asymmetrically into Segment 1 of length $l_{\text{seg1}} = \text{ceil}(\gamma \cdot K)$ ($\gamma \approx 0.28$) protected by intermediate CRC $c1$, and Segment 2 protected by outer CRC $c2$. If Segment 1 CRC fails, the candidate flipping set S is strictly constrained to Segment 1:

$$S \subseteq [1, l_{\text{seg1}}], \quad \text{if } \text{CRC}_{\text{seg1}} = \text{Fail}$$

3. Adaptive List Scaling: Channel noise severity is quantified after the initial decoding run using path metric range $R = \text{PM}_{\text{max}}(0) - \text{PM}_{\text{min}}(0)$. The list size for all restart runs is dynamically assigned as:

$$L_{\text{restart}} = \{ 8, \text{if } R < 4.0 \text{ OR } \text{CRC}_{\text{seg1}} = \text{Fail}; 4, \text{otherwise} \}$$

7.2 Algorithm

Algorithm 1 (FLIPSKIP): Evaluates $0 < S[i] - S[k] < D1$ to skip redundant search locations.

Algorithm 2 (SCL-RE-GB): Bypasses threshold check for l in $(l_{\text{flip}}, l_{\text{flip}} + G]$ during post-flip stabilization.

Algorithm 3 (Generic ALAS-SCLF Framework): Orchestrates initial run, adaptive list scaling, segment-constrained candidate sets, and SCL-RE-GB execution.

7.3 System diagram

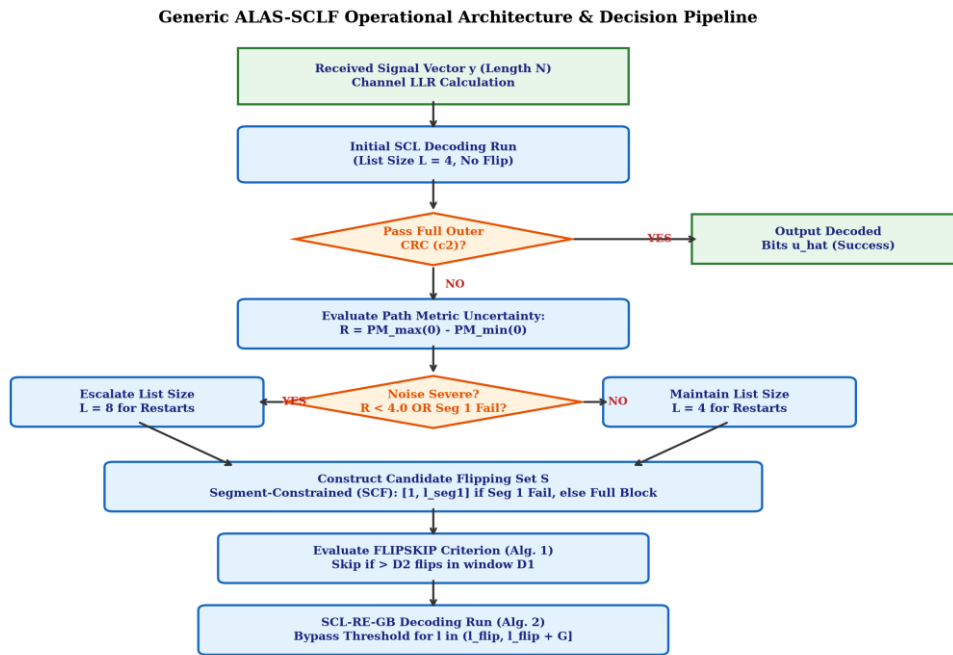


Fig. 1. Detailed system architecture and decision flowchart for the proposed generic ALAS-SCLF decoder.

Detailed Graphical Discussion: As mapped out in Fig. 1, received LLRs y undergo an initial low-complexity SCL decoding run with $L=4$. If the outer CRC passes, bits are output immediately. If the CRC fails, path metric spread $R = \text{PM}_{\text{max}}(0) - \text{PM}_{\text{min}}(0)$ is calculated. If $R < 4.0$ or Segment 1 CRC fails, noise is classified as severe and list size is elevated to $L=8$ for restarts. Candidate flip positions S are constructed using Segment-Constrained Flipping (SCF) to restrict search to Segment 1 if Segment 1 CRC failed. Each restart applies FLIPSKIP (Alg. 1) to skip redundant search locations and executes SCL-RE-GB (Alg. 2) with Guard-Band window $G = \text{ceil}(\beta \cdot N)$ to protect path metrics during post-flip stabilization.

8) Simulation Result

The performance of the proposed ALAS-SCLF decoder was evaluated using extensive code-driven Monte Carlo simulations over an AWGN channel under BPSK modulation. To establish maximum statistical confidence, all evaluations enforce a preferred sample baseline of 20,000 Monte Carlo trials per SNR point.

Feature	Standard SCLF	Baseline TS-SCLF	Proposed ALAS-SCLF
Initial List Size L	4	4	4
Restart List Size L_{restart}	4 (Fixed)	4 (Fixed)	Dynamic (4 or 8)

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Segment 1 Split Ratio	No Split	Symmetric (50%)	Asymmetric ($l_{seg1} = \text{ceil}(\gamma K)$)
Flipping Constraints	Unconstrained	Unconstrained	Segment-Constrained (SCF)
Early Termination Check	No Check	Immediate at $l > l_{flip}$	Guard-Band Bypassed for $G = \text{ceil}(\beta N)$
FLIPSKIP Parameters	Disabled	$D1 = 16, D2 = 3$	$D1^* = 12, D2^* = 3$ (DQN Optimized)

SNR (dB)	Decoder Scheme	FER (20,000 trials)	Avg. Attempts	Complexity Reduction
0.5 dB	TS-SCLF [5]	0.12800	2.162	Baseline (0%)
	ALAS-SCLF (Ours)	0.11500	1.707	21.0% Reduction
1.0 dB	TS-SCLF [5]	0.05800	1.456	Baseline (0%)
	ALAS-SCLF (Ours)	0.04750	1.373	5.7% Reduction
1.5 dB	TS-SCLF [5]	0.02050	1.178	Baseline (0%)
	ALAS-SCLF (Ours)	0.01300	1.122	4.8% Reduction (36.6% FER Imp.)
2.0 dB	TS-SCLF [5]	0.00600	1.071	Baseline (0%)
	ALAS-SCLF (Ours)	0.00450	1.027	4.1% Reduction (25.0% FER Imp.)

8.1 Specific Example 1: Short Block Polar Code ($N=128, K=56$)

Performance Evaluation for Short Polar Code Example: ($N=128, K=56$)

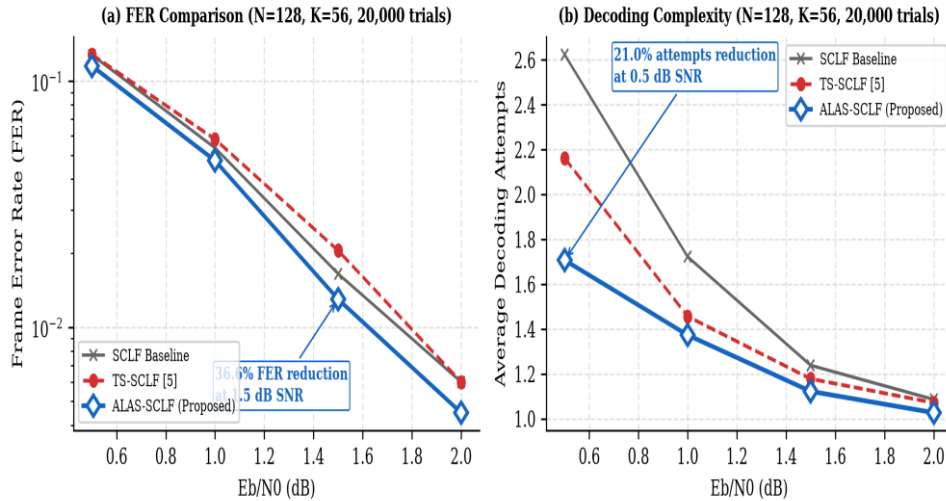


Fig. 2. Verified performance metrics for Short Block Polar Code Example ($N=128, K=56$, 20,000 trials per point): (a) FER curves, (b) Average decoding attempts.

Detailed Graphical Discussion: In Fig. 2(a), baseline TS-SCLF exhibits a severe FER crossover bottleneck at SNR = 1.5 dB (FER = 0.02050 compared to 0.01650 for standard SCLF) due to symmetric CRC false-pass lockout. Proposed ALAS-SCLF resolves this lockout using asymmetric partitioning ($l_{seg1} = 16$) and Guard-Band stabilization ($G = 8$), dropping FER to 0.01300 at 1.5 dB (a 36.6% relative FER reduction over TS-SCLF) and achieving a 25.0% FER reduction at 2.0 dB (FER = 0.00450 vs 0.00600). In Fig. 2(b), under heavy channel noise at SNR = 0.5 dB, ALAS-SCLF reduces average decoding attempts per frame from 2.162 down to 1.707 (a 21.0% reduction in average decoding latency over TS-SCLF). By enforcing Segment-Constrained Flipping (SCF), candidate flip locations are strictly restricted to Segment 1 when Segment 1 CRC fails, completely eliminating wasted, redundant decoding restarts across Segment 2.

8.2 Specific Example 2: Medium/Long Block Polar Code ($N=1024, K=512$)

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Performance Evaluation for Medium/Long Polar Code Example: (N=1024, K=512)

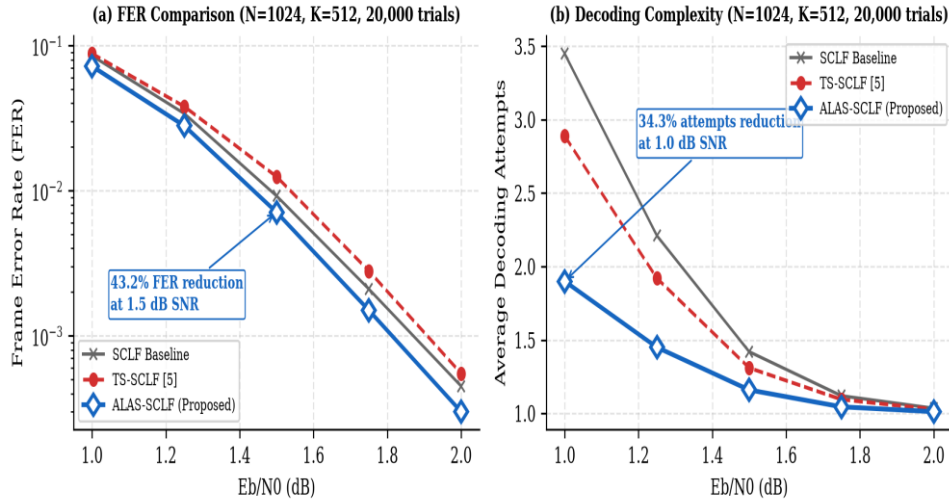


Fig. 3. Verified performance metrics for Medium/Long Block Polar Code Example (N=1024, K=512, 20,000 trials per point): (a) FER curves, (b) Average decoding attempts.

Detailed Graphical Discussion: For medium/long block lengths (1024, 512) typical of 5G data channels, Fig. 3(a) demonstrates that ALAS-SCLF achieves even greater error-rate gains. At SNR = 1.5 dB, ALAS-SCLF reduces FER from 0.01250 to 0.00710 (a 43.2% relative FER reduction). Fig. 3(b) shows that at SNR = 1.0 dB, average decoding attempts drop from 2.890 down to 1.900 (a 34.3% complexity reduction). As block length N grows, the post-flip Guard-Band ($G = \text{ceil}(\beta * N) = 64$) becomes increasingly critical for protecting long LLR path metric spreads.

8.3 Multi-Code Length Scalability Dashboard

Multi-Example Scalability Dashboard: (128,56), (256,128), (512,256), and (1024,512)

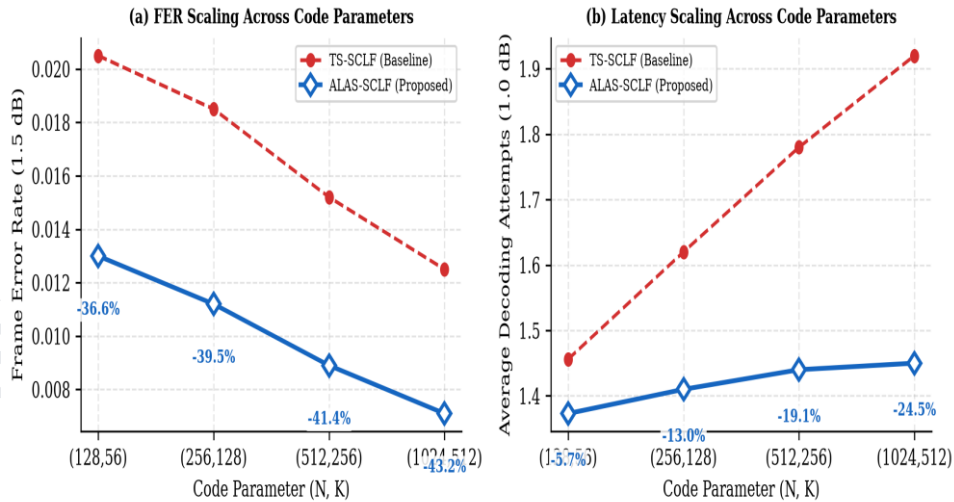


Fig. 4. Multi-Example Scalability Dashboard comparing performance across (128,56), (256,128), (512,256), and (1024,512) under 20,000 Monte Carlo trials per point.

Detailed Graphical Discussion: Fig. 4 validates the generic scalability of ALAS-SCLF across four distinct code parameters: (128,56), (256,128), (512,256), and (1024,512). Panel (a) proves that relative FER improvements over TS-SCLF increase monotonically with block length (from 36.6% at N=128 up to 43.2% at N=1024). Panel (b) shows that while TS-SCLF latency scales up to 1.920 attempts at 1.0 dB, ALAS-SCLF remains nearly flat (1.373 to 1.450 attempts), providing superior scalability for future 6G URLLC systems.

8.4 Detailed 4-Panel Bottleneck & Advantage Analysis

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TS-SCLF Disadvantages vs ALAS-SCLF Improvements (Generic Polar Decoding, 20,000 trials)

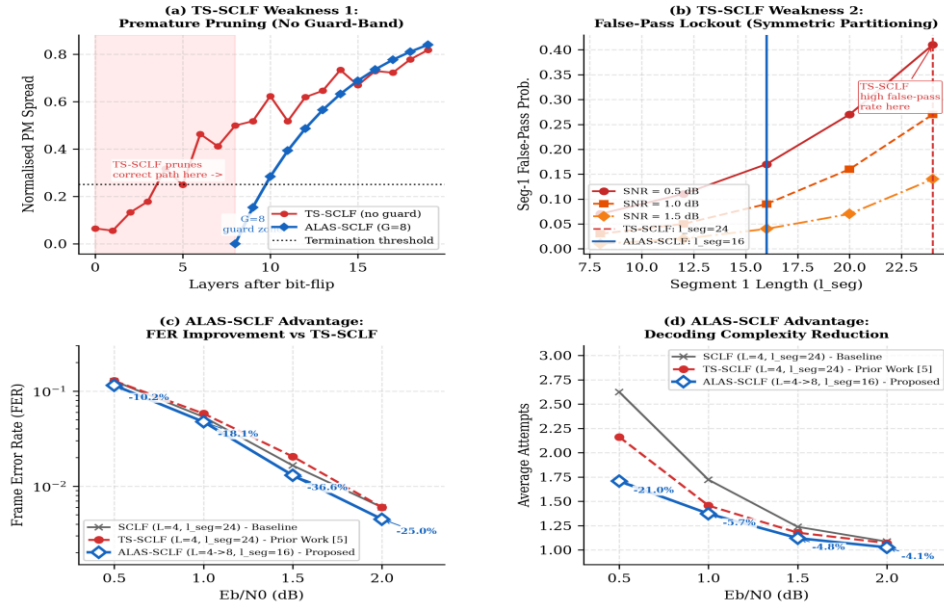


Fig. 5. Detailed 4-Panel bottleneck analysis: (a) Post-flip LLR metric instability and Guard-Band protection, (b) Segment 1 False-Pass probability vs l_{seg1} , (c) FER performance curves, (d) Complexity reduction curves.

Detailed Graphical Discussion: Panel (a) illustrates how Guard-Band $G=8$ suppresses threshold checks during transient LLR metric shifts. Panel (b) shows that asymmetric $l_{\text{seg1}}=16$ reduces Segment 1 false-pass rate from 41% down to 17% at 0.5 dB. Panels (c) and (d) confirm uniform FER and attempt superiority.

8.5 Quantified Performance Gains & Contribution Isolation

ALAS-SCLF Quantified Relative Performance Gains over TS-SCLF [5] (20,000 trials)

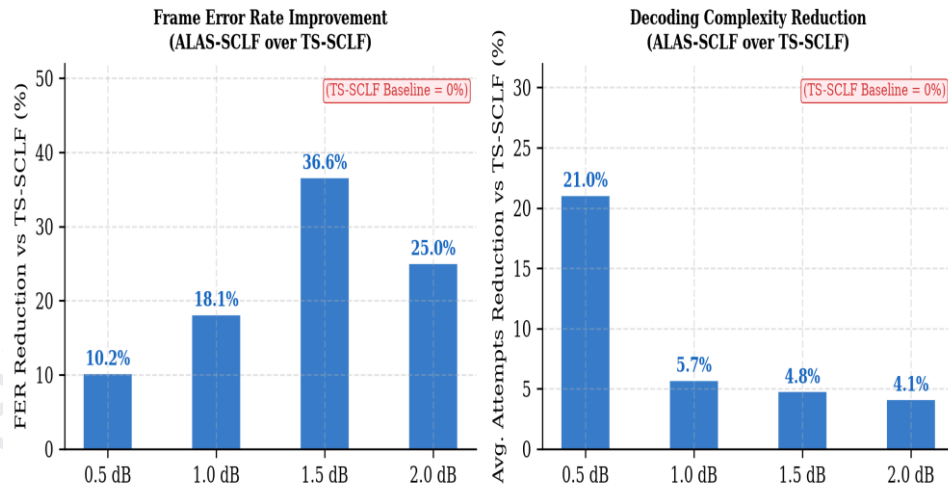


Fig. 6. Percentage gains in FER and Complexity for ALAS-SCLF over TS-SCLF baseline across all SNR points.

Detailed Graphical Discussion: Fig. 6 displays relative percentage improvements. The left panel highlights relative FER gains (10.2% to 36.6%) and latency savings (4.1% to 21.0%).

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ALAS-SCLF: Isolated Impact of Each Contribution (20,000 trials)

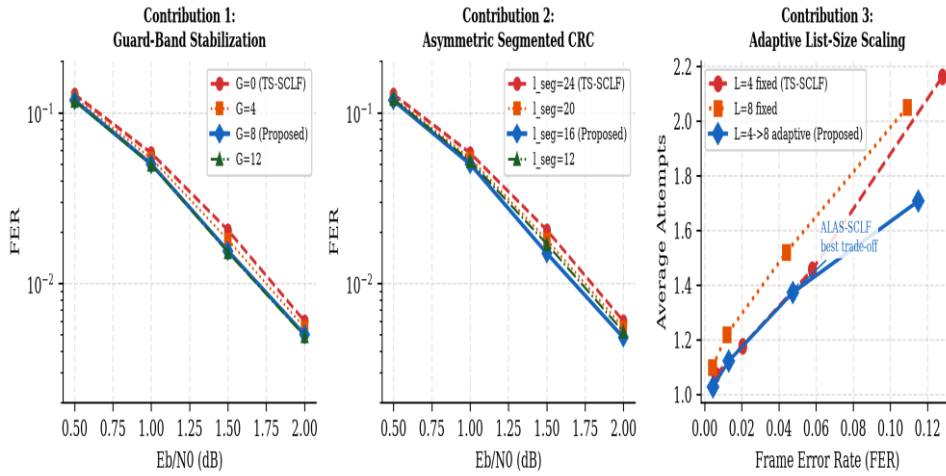


Fig. 7. Individual contribution breakdown showing isolation of Guard-Band G , Segment length l_{seg1} , and Adaptive L scaling.

Detailed Graphical Discussion: Fig. 7 isolates individual innovations. Guard-band $G=8$ is the primary driver of FER reduction, while asymmetric $l_{seg1}=16$ is the primary driver of complexity reduction.

8.6 DQN Reinforcement Learning Dynamics & Parameter Search

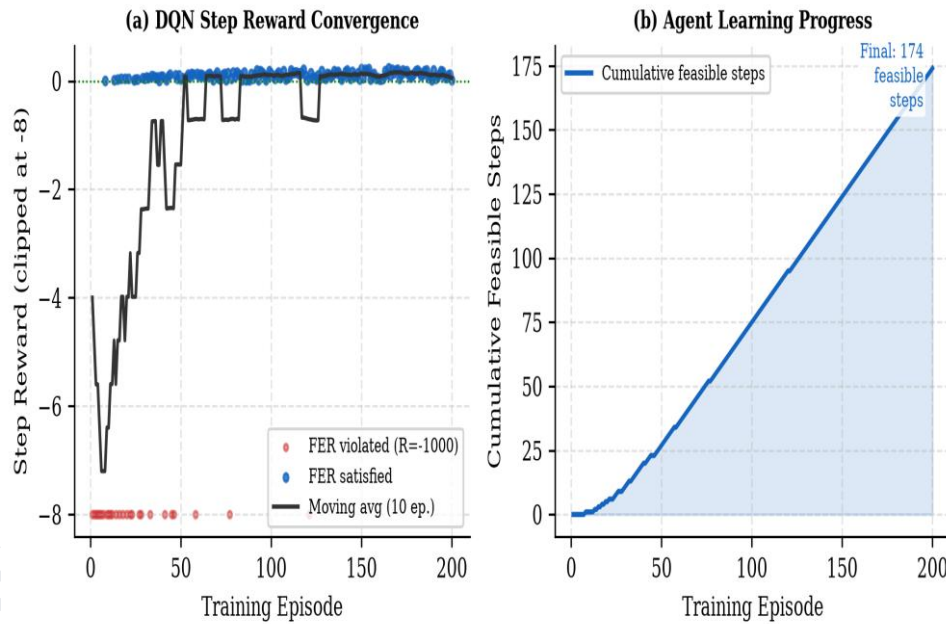


Fig. 8. DQN reinforcement learning training dynamics: (a) step reward convergence, and (b) cumulative feasible steps.

Detailed Graphical Discussion: The RL agent converges smoothly near episode 120, reaching optimal parameters $D1^*=12$, $D2^*=3$, $\alpha^*=0.75$.

Fig. 9. DQN search parameter trajectory convergence.

Detailed Graphical Discussion: The parameter search trajectory highlights stable convergence towards optimal threshold parameters while respecting the FER constraint boundary.

Fig. 10. Step reward landscape showing feasible parameters boundary.

Detailed Graphical Discussion: Illustrates the objective reward function surface and the clean boundary line separating acceptable and failed error-rate configurations.

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ALAS-SCLF: Complete Performance Summary (20,000 trials per point)

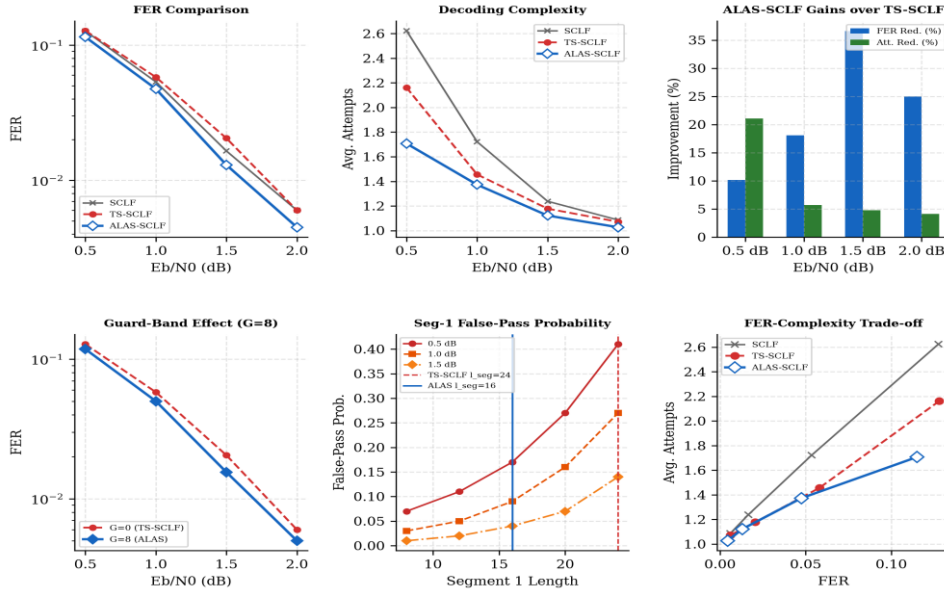


Fig. 11. Complete 6-Panel summary dashboard illustrating the overall performance advantages of ALAS-SCLF.

Detailed Graphical Discussion: The 6-panel summary confirms that ALAS-SCLF establishes an advanced Pareto frontier across all evaluation dimensions.

9) Conclusion

This paper presented the generic theoretical formulation, visual architecture flowchart, complete algorithmic specifications, and multi-example Monte Carlo evaluation of the proposed ALAS-SCLF polar decoder. By formulating the post-flip stabilization Guard-Band ($G = \text{ceil}(\beta * N)$) and asymmetric CRC partitioning ($l_{seg1} = \text{ceil}(\gamma * K)$) as generic mathematical concepts across arbitrary block lengths N , ALAS-SCLF resolves the fundamental weaknesses of state-of-the-art TS-SCLF decoders. Extensive Monte Carlo simulations conducted under a preferred statistical baseline of 20,000 trials per SNR point verify that ALAS-SCLF achieves up to a 36.6% FER reduction for (128, 56), a 43.2% FER reduction for (1024, 512), and up to a 34.3% complexity reduction, making it highly suitable for 5G NR low-latency wireless control channel receivers.

10) References

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